

Theorem 2.15 (Representation theorem for polytopes).

A subset $P \subseteq \mathbb{R}^d$ is a polytope if and only if it can be described in any of the following (equivalent) ways:

- (1) an affine projection of a simplex,
- (2) all the convex combinations of a finite point set,
- (3) all the convex combinations of the vertex set $\text{vert}(P)$,
- (4) the union of all simplices spanned by a finite set of points,
- (5) the projection of the d -skeleton of a simplex,
- (6) a bounded intersection of (closed) halfspaces,
- (7) a bounded intersection of facet-defining (closed) halfspaces, one for each facet, and of the affine hull of P .

Proof. The equivalence of (1) and (2) is from the definitions of a convex hull and a simplex. It is just an example of translation of a geometric statement (1) into an algebraic one (2). Similarly, (4) $\iff$ (5) is such a translation.

The equivalence of (2) and (3) is from Proposition 2.2, while the equivalence of (1) and (5) is from Carathéodory's theorem 1.15(ii).

The equivalence of (2) and (6) is the main theorem on polytopes, Theorem 1.1, which we proved by Fourier-Motzkin elimination. Instead, we could also argue that (2) $\iff$ (6) follows from polarity.

Finally, for (3) $\iff$ (7) we reduce this to the full-dimensional case, and then use that the facets of P^Δ correspond to the vertices of P under polarity, by Theorem 2.12 and its corollaries. In particular, the facet-defining inequalities are uniquely determined (if we write them as $\mathbf{a}_i \mathbf{x} \leq 1$), and none of them can be deleted. $\square$

2.5 Simplicial and Simple Polytopes

Proposition 2.16. For any d -dimensional polytope P , the following conditions are equivalent:

- (i) every facet of P is a simplex, i.e., P is *simplicial*,
- (ii) every proper face of P is a simplex,
- (iii) every facet has d vertices,
- (iv) every k -face has $k + 1$ vertices, for $k \leq d - 1$,
- (v) every lower interval $[\hat{0}, F] \subseteq L(P)$ in the face lattice with $F \neq \hat{1}$ is a boolean poset.

Similarly, the following conditions are equivalent:

- (i) every vertex figure of P is a simplex, i.e., P is simple,
- (ii) every iterated vertex figure of P is a simplex,
- (iii) every vertex is in d facets,
- (iv) every k -face is contained in $d - k$ facets, for $k \geq 0$,
- (v) every upper interval $[F, \hat{1}] \subseteq L(P)$ in the face lattice with $F \neq \hat{0}$ is a boolean poset.

In particular, a polytope is simplicial if and only if any combinatorially polar polytope is simple, and it is simple if and only if any combinatorially polar polytope is simplicial.

Proof. This is easy. It only uses the fact that every $(d - 1)$ -dimensional simplex has $\binom{d}{k+1}$ k -faces, and the fact that its face lattice is the boolean poset B_d of all subsets of a d -set.

The first and second parts of the theorem are equivalent, via polarity. Here we use that the opposite of a boolean poset is isomorphic to the poset itself. $\square$

For the following, assume (without loss of generality) that we consider full-dimensional polytopes. For any simplicial polytope $P = \text{conv}(V)$ we can perturb the vertex coordinates “a little” without changing the combinatorial type. From this we get a combinatorially equivalent polytope with rational vertex coordinates, which is what one calls a *rational polytope*.

Similarly, for any simple polytope $P = P(A, \mathbf{z})$ we can perturb the defining inequalities “a little” to get inequalities with rational coefficients. Clearing the denominators, we can get representations with integral vertex coordinates (as a *lattice polytope*). This proves the following result.

Proposition 2.17. *For every simple or simplicial polytope P , there is a combinatorially equivalent polytope $P' \simeq P$ with integral vertex coordinates.*

This answers one question, and opens up two new ones: first, is this true for all polytopes? We will see later that it holds for polytopes of dimension $d \leq 3$ (the case $d \leq 2$ is trivial), but it fails in general (see Lecture 6). Second, if integral coordinates exist, can we keep them reasonably small? Again, the answer is yes if we are in low dimension, but in general we have to cope with coordinates that grow doubly exponential in terms of the number of vertices; see Goodman, Pollack & Sturmfels [238]. For fixed dimension $d = 3$, however, the problem seems to be open (Problem 4.16*).

We already saw numerous examples of simple and simplicial polytopes in Lecture 0. The d -simplex, the d -cubes, and the dodecahedron are simple polytopes. The d -simplex, the octahedron, the icosahedron, and all cyclic

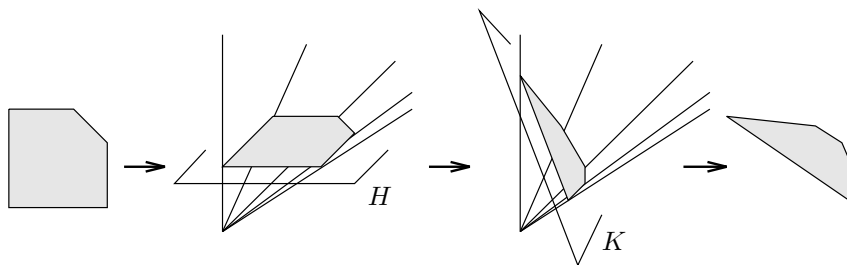
polytopes are examples of simplicial polytopes. Note that in their “usual” symmetric version as regular polytopes, the dodecahedron and the icosahedron do *not* have rational coordinates. (In fact, as regular polytopes they cannot be represented in rational coordinates!)

Any d -polytope that is both simple and simplicial is a simplex if $d \geq 3$ — this was covered in Exercise 0.0. To see this, let P be simple, and consider a vertex $\mathbf{v}$ of P . This vertex is in exactly d facets, which are all simplices. Looking at the vertex figure, we see that $\mathbf{v}$ is on d edges, and the d vertices $\mathbf{v}_1, \dots, \mathbf{v}_d$ adjacent to $\mathbf{v}$ are also adjacent to each other: here we use the condition $d \geq 3$. This means, since the same argument could start at $\mathbf{v}_i$, that there are no other vertices than $\mathbf{v}, \mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_d$ adjacent to $\mathbf{v}_i$. Thus the vertex set of P is $\{\mathbf{v}, \mathbf{v}_1, \dots, \mathbf{v}_d\}$, and P is the simplex spanned by this set.

2.6 Appendix: Projective Transformations

Although linear transformations are our main tool to “put polytopes where we need them,” it is sometimes convenient to use more general transformations, which allow us even to “adjust the shape of a given polytope,” known as *projective transformations*.

We can describe projective transformations in a very simple way with the tools of this lecture (in particular, without construction of projective space and use of projective geometry). For this, we proceed as follows. Given a polytope $P \subseteq \mathbb{R}^d$, we embed it into an affine hyperplane $H \subseteq \mathbb{R}^{d+1}$, and construct $\text{homog}(P)$, the homogenization of P . By construction, this is a pointed cone. Now we cut this cone by a different hyperplane $K \subseteq \mathbb{R}^{d+1}$, which is then identified with $\mathbb{R}^d$ by an affine map.



Here K is required to be an *admissible hyperplane*: an affine hyperplane that intersects every ray in $\text{homog}(P)$ that starts at $\mathbf{0}$. Under this condition, we get a new polytope $P' \subseteq \mathbb{R}^d$, which is affinely isomorphic to $K \cap \text{homog}(P)$, and thus combinatorially equivalent to P :

$$L(P) \cong L(\text{homog}(P)) \cong L(P').$$